

Marco Ciccone

PERSONAL INFORMATION

Surname Name: Ciccone Marco
Nationality: Italian
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RESEARCH INTERESTS

My research vision is to enable AI agents to learn how to solve new tasks by reusing knowledge extracted from solving previous ones. **Transfer learning** is a central component of human intelligence, allowing us to recognize patterns, decompose complex tasks, and re-apply previous solutions to new problems. My research aims to bridge the gap between artificial and human intelligence by endowing AI agents with *modularity* and *plasticity*, which are crucial for efficiency and deployment in real-world applications. I tackle this ambitious goal from two key perspectives:

1. **Continual Knowledge Integration**, models should incorporate or edit knowledge without forgetting previously acquired skills, unless on purpose.
2. **Collaborative Learning and Personalization**, which allows for distributed learning while preserving privacy and enabling access to the knowledge of multiple collaborative sources.

My research often has practical applications in mind, such as learning without, with limited, or weak annotations.

ACADEMIC EXPERIENCE

University College London, UK

Honorary Research Fellow

January 2022 - now

Politecnico di Torino, Italy

Postdoctoral Researcher - ELLIS PostDoc Program

June 2021 - now

Advisor: Prof. Barbara Caputo (Politecnico di Torino)

Co-Advisor: Prof. Carlo Ciliberto (University College London)

WORK EXPERIENCE

NVIDIA, Santa Clara, California

Research Intern

August 2018 - November 2018

NNAISENSE, Lugano, Switzerland

Research Intern

July 2017 - January 2018

Horus Technology, Milan, Italy

Machine Learning Engineer

June 2016 - June 2017

Projest S.p.A., Prato, Italy

IT Consultant - Software Engineer

Nov 2009 - Oct 2012

EDUCATION

Politecnico di Milano, Italy

PhD in Information Technology

April 2021

- Thesis: *“On Iterative and Conditional Computation for Visual Representation Learning”*
- Advisor: Prof. Matteo Matteucci
- Co-Advisor: PhD Jonathan Masci
- Grade: *cum Laude*

Romanian Institute of Science and Technology, Cluj-Napoca, Romania

Visiting PhD student, advised by PhD Luigi Malago’

May-July 2018

“Information Geometry of Adversarial Robustness of Neural Networks”

Politecnico di Milano, Italy

MSc in Computer Science and Engineering

April 2016

- Thesis: *“Pixelwise Semantic Segmentation of urban scenes using Recurrent Deep Neural Networks”*
- Advisors: Prof. Matteo Matteucci, PhD Francesco Visin
- Language: English
- Grade: *110/110 cum Laude*
- GPA: *28.85/30*

Università degli Studi di Firenze, Italy

BSc in Computer Engineering

July 2013

- Thesis: *“Blind Source Separation methods for multichannel audio signals based on Independent Component Analysis”*
- Advisor: Prof. Fabrizio Argenti
- Language: Italian
- Grade: *97/110*

PROFESSIONAL ACTIVITIES

Chair and Organizer

- **Competition chair NeurIPS 2023**
(co-chair with Tao Qin and Jake Albrecht)
- **Competition chair NeurIPS 2022**
(co-chair with Gustavo Stolovitzky and Jake Albrecht)
- **Demonstration and Competition chair NeurIPS 2021**
(co-chair with Barbara Caputo and Douwe Kiela)

Reviewer

- Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- AAAI Conference on Artificial Intelligence (AAAI)
- International Journal of Computer Vision (IJCV)
- International Conference on Robotics and Automation (ICRA)
- International Conference on Intelligent Robots and Systems (IROS)
- Geometric meets Deep Learning Workshop (ICCV)
- Neural Computing and Applications Journal (NCAA)

ATTENDED
SCHOOLS/EVENTS

Transylvanian Machine Learning Summer School 2018, Cluj, Romania
Acceptance Rate: 8% **16-22 July 2018**

Machine Learning Summer School 2017, Tübingen, Germany
Acceptance Rate: 13% **19-30 June 2017**

International Computer Vision Summer School 2017, Sicily, Italy
Acceptance Rate: 35% **9-15 July 2017**

Self-Organizing Conference on Machine learning 2018, Toronto, Canada
Selected on application **30 Nov - 1 Dec 2018**

TEACHING

Mediterranean Machine Learning School (M2L) 2022

Lab Tutor
Colab on Graph Neural Networks - Tutorials 2022 **September 2022**

Mediterranean Machine Learning School (M2L) 2021

Lab Tutor
Colab on Variational Auto-Encoders - Tutorials 2021 **January 2021**

Deep Learning 2017/2018, Prof. Matteo Matteucci
PhD and MSc Course at Politecnico di Milano
Teaching Assistant (PyTorch and Tensorflow Tutorials) **February 2018**

Cognitive Robotics 2016/2017, Prof. Matteo Matteucci
MSc Course at Politecnico di Milano
Teaching Assistant (Fundamentals of Deep learning) **June 2017**

PUBLICATIONS

- [1] R. Wang*, **M. Ciccone***, G. Luise, M. Pontil, A. Yapp, and C. Ciliberto, “Schedule-robust online continual learning,” *in review*, 2022
- [2] D. Shenaj*, E. Fani*, M. Toldo, D. Caldarola, A. Tavera, U. Michieli†, **M. Ciccone**†, P. Zanuttigh†, and B. Caputo†, “Learning across domains and devices: Style-driven source-free domain adaptation in clustered federated learning,” in *2023 IEEE Winter Conference on Applications of Computer Vision (WACV)*, IEEE, 2023
- [3] D. Caldarola*, B. Caputo, and **M. Ciccone***, “Improving generalization in federated learning by seeking flat minima,” in *European Conference on Computer Vision*, 2022
- [4] L. Fantauzzo, E. Fani, D. Caldarola, A. Tavera, F. Cermelli, **M. Ciccone**, and B. Caputo, “Feddrive: Generalizing federated learning to semantic segmentation in autonomous driving,” in *Proceedings of the 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2022
- [5] R. Zacccone, A. Rizzardi, D. Caldarola, **M. Ciccone**, and B. Caputo, “Speeding up heterogeneous federated learning with sequentially trained superclients,” in *26th International Conference on Pattern Recognition (ICPR)*, IEEE, 2022
- [6] L. Carminati, F. Cacciamani, **M. Ciccone**, and N. Gatti, “A marriage between adversarial team games and 2-player games: Enabling abstractions, no-regret learning, and subgame solving,” in *Proceedings of the 39th International Conference on Machine Learning*, Proceedings of Machine Learning Research, PMLR, 17–23 Jul 2022

- [7] F. Cermelli*, D. Fontanel*, A. Tavera*, **M. Ciccone**, and B. Caputo, “Incremental learning in semantic segmentation from image labels,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022
- [8] L. Carminati, F. Cacciamani, **M. Ciccone**, and N. Gatti, “Public information representation for adversarial team games,” in *NeurIPS 2021 Workshop Cooperative AI*, 2021 **Oral and Best paper award, CooperativeAI Workshop NeurIPS 2021**
- [9] M. Planamente*, C. Plizzari*, M. Cannici*, **M. Ciccone**, F. Strada, A. Bottino, M. Matteucci, and B. Caputo, “Da4event: Towards bridging the sim-to-real gap for event cameras using domain adaptation,” *IEEE Robotics and Automation Letters*, vol. 6, no. 4, pp. 6616–6623, 2021
- [10] D. Caldarola, M. Mancini, F. Galasso, **M. Ciccone**, E. Rodola, and B. Caputo, “Cluster-driven graph federated learning over multiple domains,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 2749–2758, June 2021
- [11] M. Cannici, C. Plizzari, M. Planamente, **M. Ciccone**, A. Bottino, B. Caputo, and M. Matteucci, “N-rod: A neuromorphic dataset for synthetic-to-real domain adaptation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 1342–1347, June 2021
- [12] F. Cacciamani, A. Celli, **M. Ciccone**, and N. Gatti, “Multi-agent coordination in adversarial environments through signal mediated strategies,” in *Proceedings of the 20th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2021
- [13] M. Cannici, **M. Ciccone**, A. Romanoni, and M. Matteucci, “A differentiable recurrent surface for asynchronous event-based data,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 1–17, Springer, 2020
- [14] M. Cannici, **M. Ciccone**, A. Romanoni, and M. Matteucci, “Asynchronous convolutional networks for object detection in neuromorphic cameras,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2019 **Best paper award, Second International Workshop on Event-based Vision and Smart Cameras Workshop at CVPR2019**
- [15] M. Cannici, **M. Ciccone**, A. Romanoni, and M. Matteucci, “Attention mechanisms for object recognition with event-based cameras,” *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2019
- [16] **M. Ciccone***, M. Gallieri*, J. Masci, C. Osendorfer, and F. Gomez, “Nais-net: Stable deep networks from non-autonomous differential equations,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018
- [17] F. Lattari*, **M. Ciccone***, M. Matteucci, J. Masci, and F. Visin, “Reconvnet: Video object segmentation with spatio-temporal features modulation,” *The 2018 DAVIS Challenge on Video Object Segmentation - CVPR Workshops*, 2018
- [18] A. Romanoni, **M. Ciccone**, F. Visin, and M. Matteucci, “Multi-view stereo with single-view semantic mesh refinement,” in *Proceedings of the IEEE International Conference on Computer Vision Workshops (ICCVW)*, pp. 706–715, 2017
- [19] F. Visin, **M. Ciccone**, A. Romero, K. Kastner, C. Kyunghyun, Y. Bengio, M. Matteucci, and A. Courville, “ReSeg : A Recurrent Neural Network-based Model for Semantic Segmentation,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2016 **Best paper award, DeepVision Workshop at CVPR2016**